

CommonHealth — Concept Note

Integrated Longevity Research Platform — EIC Pathfinder Open 2026

Field	Value
Coordinator	NGO Georgia Longevity Alliance (Reg. №404506520, ს.ს.ო.პ / non-profit)
Country	Georgia (Horizon Europe Associated Country since 2023)
Principal Investigator	Dr. Jaba Tkemaladze, MD — 150+ publications, Editor-in-Chief <i>Longevity Horizon</i>
Budget	€3.0M
Duration	36 months
Call	EIC Pathfinder Open 2026 — deadline 12 May 2026
Contact	jaba@longevity.ge · +995 555 185 161 (Tbilisi, GMT+4)

Vision

To transition longevity research from fragmented, descriptive, single-mechanism studies to an **integrated, systems-level science of aging** — enabled by privacy-preserving federated infrastructure that unifies multi-modal data streams (molecular, physiological, clinical, traditional) without centralizing sensitive data.

Breakthrough Claim

CommonHealth delivers the first **theory-to-bedside pipeline** that integrates:

- A novel mechanistic theory (CDATA — Centriolar Damage Accumulation Theory of Aging)
- A falsifiable physiological biomarker (χ_{Ze} from Ze Vectors Theory)
- Open-source federated learning infrastructure (FCLC) meeting EU AI Act Article 22 compliance
- Wearable biomarker hardware (BioSense, TRL 1→3)
- A real-world clinical pilot (Aqtivirebuli — traditional Georgian nutritional intervention for iron-deficiency anemia)

Work Packages

WP	Title	Budget	Duration	TRL
WP1	FCLC Federated learning infrastructure with DP $\epsilon \leq 1.0$, SecAgg+	— €0.6M	12 mo	3→5
WP2	Ze — χ _Ze biomarker validation (MPI- LEMON, Dortmund, Cuban N=196)	€0.5M	12 mo	2→4
WP3	CDATA — Centriolar aging Phase 0 (TTLL6, CCP1, serial BMT mice)	€0.8M	24 mo	2→4
WP4	BioSense — EEG+HRV wearable prototype with onboard χ _Ze	€0.6M	18 mo	1→3
WP5	Aqtivirebuli — Clinical pilot N=50 anemic patients, 8 weeks	€0.5M	12 mo	4→6

Partners (current status 2026-04-17)

- **Coordinator confirmed:** NGO Georgia Longevity Alliance (host)
- **Industry confirmed:** BioViva (Liz Parrish) — CDATA WP3
- **Clinical partner confirmed:** Dr. Shashviashvili (Georgia) — WP5
- **EU technical partner:** In recruitment (Fraunhofer IESE / KU Leuven COSIC — LoI requested)
- **HSC experimental collaborator:** Prof. Hartmut Geiger (Ulm) — letter sent

Impact Pathways

Scientific: First unified mechanistic theory + biomarker + federated data architecture for aging research. **Technological:** Open-source FCLC platform deployable across EU health systems; wearable BioSense prototype. **Societal:** Democratized longevity research; anemia intervention model scalable to 2 billion affected globally. **Economic:** Spinoff potential in medical devices (BioSense) and functional foods (Aqtivirebuli); Georgia positioned as Horizon Europe longevity hub.

Honest Disclosures

- PI does not hold PhD (mitigated: 150 publications, Editor-in-Chief role, Georgia's eligibility)
- Structural COI: PI = President of Host (full mitigation in separate COI Declaration with Supervisory Board + annual external audit)
- χ_{Ze} is research-stage, NOT clinically validated (Phase 1 validation in WP2)
- CDATA BHCA score: 8/27 currently (Class WEAK) — Phase 0 WP3 designed to upgrade to $\geq 18/27$

Full Part A + Part B drafts available on request. Letters of Support required by 25 April 2026.